

Assignment 4: Optimizing Transformer Translation with Ray Tune & Optuna

Tejas Gupta
Roll No: B22CS093

1 Baseline Metrics (100 Epochs)

The baseline model was trained using the provided `en_to_hi.ipynb` notebook without any modifications. The architecture is a 6-layer Transformer with `d_model=512`, `num_heads=8`, `d_ff=2048`, `dropout=0.1`, `batch_size=60`, and `lr=1e-4`.

Metric	Value
Epochs	100
Final Training Loss	~0.35
BLEU Score (NLTK)	52.50

Table 1: Baseline performance on the English-Hindi translation task.

2 Hyperparameter Search Configuration

Five hyperparameters were selected for tuning using Ray Tune with OptunaSearch and the ASHA scheduler for early stopping of underperforming trials.

Hyperparameter	Search Type	Range
Learning Rate (<code>lr</code>)	Log-uniform	$[1 \times 10^{-5}, 1 \times 10^{-3}]$
Batch Size	Categorical	{16, 32, 64}
Attention Heads (<code>num_heads</code>)	Categorical	{4, 8}
Feed-Forward Dim (<code>d_ff</code>)	Categorical	{1024, 2048}
Dropout Rate	Uniform	[0.1, 0.4]

Table 2: Hyperparameter search space definition.

The ASHA scheduler was configured with `max_t=30`, `grace_period=5`, and `reduction_factor=2` to aggressively prune poor trials. Five sample configurations were evaluated during the sweep.

3 Best Configuration Found

Key observations: the optimal learning rate (1.578×10^{-4}) is close to the baseline (1×10^{-4}) but slightly higher, enabling faster convergence. The higher dropout (0.272 vs. 0.1) provides better regularization, and a larger batch size (64 vs. 60) stabilizes gradient estimates.

Hyperparameter	Best Value
Learning Rate	1.578×10^{-4}
Batch Size	64
Attention Heads	8
Feed-Forward Dimension	2048
Dropout Rate	0.272
d_model (fixed)	512
Encoder/Decoder Layers (fixed)	6

Table 3: Best hyperparameter configuration selected by Optuna.

4 Final Results

The best configuration was trained for two scenarios: a full 100-epoch run and a 30-epoch efficiency run.

Metric	Baseline	Best (100 ep.)	Efficiency (30 ep.)
BLEU Score	52.499	59.44	53.95
Final Loss	~ 0.35	0.1818	—
Epochs	100	100	30
Training Time	—	6529.6 s	~ 1962 s

Table 4: Comparison of baseline, best-tuned, and efficiency-tuned models.

Efficiency Goal

The optimized model achieved a BLEU score of **53.95** in only **30 epochs**, exceeding the baseline BLEU of 52.50. This demonstrates that the tuned hyperparameters allow the model to converge to competitive performance in roughly one-third of the original training budget.

When trained for the full 100 epochs, the tuned configuration reaches a BLEU score of **59.44**, a +6.94 point improvement over the baseline, with a significantly lower final loss (0.1818 vs. ~ 0.35).

5 Conclusion

Hyperparameter optimization via Ray Tune with OptunaSearch and the ASHA early-stopping scheduler successfully identified a configuration that:

- Matches baseline BLEU in 30 epochs instead of 100 ($3.3\times$ speedup).
- Exceeds baseline BLEU by +6.94 points when given the same 100-epoch budget.
- Achieves substantially lower training loss, indicating better model fit.

The saved model weights are available as `B22CS093_ass_4_best_model.pth` (100-epoch best) and `B22CS093_ass_4_efficiency_model.pth` (30-epoch efficient) on Hugging Face: <https://huggingface.co/Tron2703/Transformer-English-Hindi-B22CS093>.